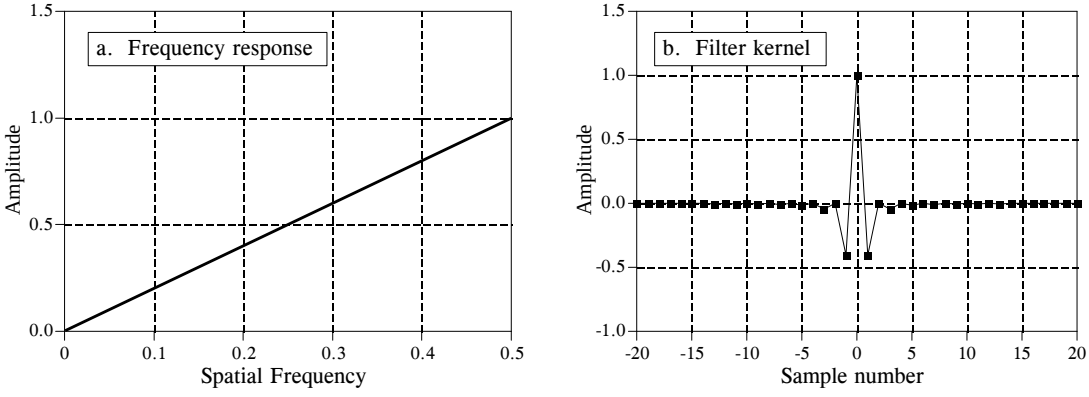

FIGURE 25-19
Backprojection filter. The frequency response of the backprojection filter is shown in (a), and the corresponding filter kernel is shown in (b). Equation 25-2 provides the values for the filter kernel.

This "radial to rectangular" conversion is also the key for understanding filtered backprojection. The radial arrangement is the spectrum of the backprojected image, while the rectangular grid is the spectrum of the correct image. If we compare one small region of the radial spectrum with the corresponding region of the rectangular grid, we find that the sample values are identical. However, they have a different *sample density*. The correct spectrum has uniformly spaced points throughout, as shown by the even spacing of the rectangular grid. In comparison, the backprojected spectrum has a higher sample density near the center because of its radial arrangement. In other words, the spokes of a wheel are closer together near the hub. This issue does not affect Fourier reconstruction because the interpolation is from the *values* of the nearest neighbors, not their *density*.

The filter in filtered backprojection cancels this unequal sample density. In particular, the frequency response of the filter must be the *inverse* of the sample density. Since the backprojected spectrum has a density of $1/f$, the appropriate filter has a frequency response of $H[f] = f$. This frequency response is shown in Fig. 25-19a. The filter kernel is then found by taking the inverse Fourier transform, as shown in (b). Mathematically, the filter kernel is given by:

EQUATION 25-2
The filter kernel for filtered backprojection. Figure 25-19b shows a graph of this kernel.

$$h[0] = 1$$

$$h[k] = 0 \qquad \text{for even values of } k$$

$$h[k] = \frac{4/\pi^2}{k^2} \qquad \text{for odd values of } k$$